MSG No. 196.6 F_S23 [Type IV, cubic]

Table 1: Wyckoff site: `8a`, site symmetry: `23.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[25,26,27,28,29,30,31,32,33,34,35,36]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[37,38,39,40,41,42,43,44,45,46,47,48]`
5	$\left[0, 0, \frac{1}{2}\right]$	`[49,50,51,52,53,54,55,56,57,58,59,60]`
6	$\left[0, \frac{1}{2}, 0\right]$	`[61,62,63,64,65,66,67,68,69,70,71,72]`
7	$\left[\frac{1}{2}, 0, 0\right]$	`[73,74,75,76,77,78,79,80,81,82,83,84]`
8	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[85,86,87,88,89,90,91,92,93,94,95,96]`

Table 2: Wyckoff site: `8b`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,5,6,14,22,23,27,31,36,40,44,45]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[2,10,11,13,17,18,28,32,33,39,43,48]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,7,12,16,20,21,25,29,30,38,46,47]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[4,8,9,15,19,24,26,34,35,37,41,42]`
5	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[49,53,54,62,70,71,75,79,84,88,92,93]`
6	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[50,58,59,61,65,66,76,80,81,87,91,96]`
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[51,55,60,64,68,69,73,77,78,86,94,95]`
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[52,56,57,63,67,72,74,82,83,85,89,90]`

Table 3: Wyckoff site: `24c`, site symmetry: `22'2'..`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	`[1,14,51,64]`
2	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	`[2,13,52,63]`
3	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[3,16,49,62]`
4	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[4,15,50,61]`
5	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[5,36,58,80]`
6	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[6,45,71,79]`
7	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[7,47,69,78]`
8	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[8,34,60,77]`
9	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[9,42,67,83]`
10	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[10,32,53,84]`
11	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[11,43,66,81]`
12	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[12,29,56,82]`
13	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[17,48,70,92]`
14	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[18,33,59,91]`
15	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[19,35,57,90]`
16	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[20,46,72,89]`
17	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[21,30,55,95]`
18	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[22,44,65,96]`

continued ...

Table 3

No.	position	mapping
19	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[23,31,54,93]
20	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[24,41,68,94]
21	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[25,38,75,88]
22	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[26,37,76,87]
23	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[27,40,73,86]
24	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[28,39,74,85]

Table 4: Wyckoff site: 24d, site symmetry: 22'2'..

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	[1,2,75,76]
2	$\left[\frac{3}{4}, 0, 0\right]$	[3,4,73,74]
3	$\left[0, \frac{1}{4}, 0\right]$	[5,12,68,70]
4	$\left[0, 0, \frac{1}{4}\right]$	[6,9,55,59]
5	$\left[0, 0, \frac{3}{4}\right]$	[7,11,54,57]
6	$\left[0, \frac{3}{4}, 0\right]$	[8,10,65,72]
7	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	[13,14,87,88]
8	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	[15,16,85,86]
9	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[17,24,56,58]
10	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[18,21,67,71]
11	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[19,23,66,69]
12	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[20,22,53,60]
13	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[25,26,51,52]
14	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[27,28,49,50]
15	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$	[29,36,92,94]
16	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[30,33,79,83]
17	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[31,35,78,81]
18	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	[32,34,89,96]
19	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[37,38,63,64]
20	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[39,40,61,62]
21	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[41,48,80,82]
22	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[42,45,91,95]
23	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[43,47,90,93]
24	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[44,46,77,84]

Table 5: Wyckoff site: 32e, site symmetry: .3.

No.	position	mapping
1	$[x, x, x]$	[1,5,6]
2	$[x, -x, -x]$	[2,10,11]
3	$[-x, x, -x]$	[3,7,12]
4	$[-x, -x, x]$	[4,8,9]
5	$\left[x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[13,17,18]
6	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[14,22,23]

continued ...

Table 5

No.	position	mapping
7	$\left[-x,\, x+\frac{1}{2},\, \frac{1}{2}-x\right]$	[15,19,24]
8	$\left[-x,\, \frac{1}{2}-x,\, x+\frac{1}{2}\right]$	[16,20,21]
9	$\left[x+\frac{1}{2},\, x,\, x+\frac{1}{2}\right]$	[25,29,30]
10	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}-x\right]$	[26,34,35]
11	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}-x\right]$	[27,31,36]
12	$\left[\frac{1}{2}-x,\, -x,\, x+\frac{1}{2}\right]$	[28,32,33]
13	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, x\right]$	[37,41,42]
14	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, -x\right]$	[38,46,47]
15	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, -x\right]$	[39,43,48]
16	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, x\right]$	[40,44,45]
17	$\left[x,\, x,\, x+\frac{1}{2}\right]$	[49,53,54]
18	$\left[x,\, -x,\, \frac{1}{2}-x\right]$	[50,58,59]
19	$\left[-x,\, x,\, \frac{1}{2}-x\right]$	[51,55,60]
20	$\left[-x,\, -x,\, x+\frac{1}{2}\right]$	[52,56,57]
21	$\left[x,\, x+\frac{1}{2},\, x\right]$	[61,65,66]
22	$\left[x,\, \frac{1}{2}-x,\, -x\right]$	[62,70,71]
23	$\left[-x,\, x+\frac{1}{2},\, -x\right]$	[63,67,72]
24	$\left[-x,\, \frac{1}{2}-x,\, x\right]$	[64,68,69]
25	$\left[x+\frac{1}{2},\, x,\, x\right]$	[73,77,78]
26	$\left[x+\frac{1}{2},\, -x,\, -x\right]$	[74,82,83]
27	$\left[\frac{1}{2}-x,\, x,\, -x\right]$	[75,79,84]
28	$\left[\frac{1}{2}-x,\, -x,\, x\right]$	[76,80,81]
29	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, x+\frac{1}{2}\right]$	[85,89,90]
30	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-x\right]$	[86,94,95]
31	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{2}-x\right]$	[87,91,96]
32	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, x+\frac{1}{2}\right]$	[88,92,93]

Table 6: Wyckoff site: 48f, site symmetry: 2..

No.	position	mapping
1	$[x,\, 0,\, 0]$	[1,2]
2	$[-x,\, 0,\, 0]$	[3,4]
3	$[0,\, x,\, 0]$	[5,12]
4	$[0,\, 0,\, x]$	[6,9]
5	$[0,\, 0,\, -x]$	[7,11]
6	$[0,\, -x,\, 0]$	[8,10]
7	$\left[x,\, \frac{1}{2},\, \frac{1}{2}\right]$	[13,14]
8	$\left[-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	[15,16]
9	$\left[0,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[17,24]
10	$\left[0,\, \frac{1}{2},\, x+\frac{1}{2}\right]$	[18,21]
11	$\left[0,\, \frac{1}{2},\, \frac{1}{2}-x\right]$	[19,23]
12	$\left[0,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[20,22]
13	$\left[x+\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	[25,26]
14	$\left[\frac{1}{2}-x,\, 0,\, \frac{1}{2}\right]$	[27,28]
15	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[29,36]

continued ...

Table 6

No.	position	mapping
16	$\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$	[30,33]
17	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$	[31,35]
18	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[32,34]
19	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[37,38]
20	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[39,40]
21	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[41,48]
22	$\left[\frac{1}{2},\ \frac{1}{2},\ x\right]$	[42,45]
23	$\left[\frac{1}{2},\ \frac{1}{2},\ -x\right]$	[43,47]
24	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[44,46]
25	$\left[x,\ 0,\ \frac{1}{2}\right]$	[49,50]
26	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[51,52]
27	$\left[0,\ x,\ \frac{1}{2}\right]$	[53,60]
28	$\left[0,\ 0,\ x+\frac{1}{2}\right]$	[54,57]
29	$\left[0,\ 0,\ \frac{1}{2}-x\right]$	[55,59]
30	$\left[0,\ -x,\ \frac{1}{2}\right]$	[56,58]
31	$\left[x,\ \frac{1}{2},\ 0\right]$	[61,62]
32	$\left[-x,\ \frac{1}{2},\ 0\right]$	[63,64]
33	$\left[0,\ x+\frac{1}{2},\ 0\right]$	[65,72]
34	$\left[0,\ \frac{1}{2},\ x\right]$	[66,69]
35	$\left[0,\ \frac{1}{2},\ -x\right]$	[67,71]
36	$\left[0,\ \frac{1}{2}-x,\ 0\right]$	[68,70]
37	$\left[x+\frac{1}{2},\ 0,\ 0\right]$	[73,74]
38	$\left[\frac{1}{2}-x,\ 0,\ 0\right]$	[75,76]
39	$\left[\frac{1}{2},\ x,\ 0\right]$	[77,84]
40	$\left[\frac{1}{2},\ 0,\ x\right]$	[78,81]
41	$\left[\frac{1}{2},\ 0,\ -x\right]$	[79,83]
42	$\left[\frac{1}{2},\ -x,\ 0\right]$	[80,82]
43	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[85,86]
44	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[87,88]
45	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[89,96]
46	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[90,93]
47	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[91,95]
48	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[92,94]

Table 7: Wyckoff site: 48g, site symmetry: 2'..

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,50]
2	$\left[x,\ 0,\ \frac{3}{4}\right]$	[2,49]
3	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[3,52]
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[4,51]
5	$\left[\frac{1}{4},\ x,\ 0\right]$	[5,84]
6	$\left[0,\ \frac{1}{4},\ x\right]$	[6,69]
7	$\left[0,\ \frac{1}{4},\ -x\right]$	[7,71]
8	$\left[\frac{3}{4},\ -x,\ 0\right]$	[8,82]

continued ...

Table 7

No.	position	mapping
9	$\left[0,\ \frac{3}{4},\ x\right]$	[9,66]
10	$\left[\frac{1}{4},\ -x,\ 0\right]$	[10,80]
11	$\left[0,\ \frac{3}{4},\ -x\right]$	[11,67]
12	$\left[\frac{3}{4},\ x,\ 0\right]$	[12,77]
13	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[13,62]
14	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[14,61]
15	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[15,64]
16	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[16,63]
17	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[17,96]
18	$\left[0,\ \frac{3}{4},\ x+\frac{1}{2}\right]$	[18,57]
19	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-x\right]$	[19,59]
20	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[20,94]
21	$\left[0,\ \frac{1}{4},\ x+\frac{1}{2}\right]$	[21,54]
22	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[22,92]
23	$\left[0,\ \frac{1}{4},\ \frac{1}{2}-x\right]$	[23,55]
24	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[24,89]
25	$\left[x+\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	[25,74]
26	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[26,73]
27	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{4}\right]$	[27,76]
28	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	[28,75]
29	$\left[\frac{3}{4},\ x,\ \frac{1}{2}\right]$	[29,60]
30	$\left[\frac{1}{2},\ \frac{1}{4},\ x+\frac{1}{2}\right]$	[30,93]
31	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	[31,95]
32	$\left[\frac{1}{4},\ -x,\ \frac{1}{2}\right]$	[32,58]
33	$\left[\frac{1}{2},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	[33,90]
34	$\left[\frac{3}{4},\ -x,\ \frac{1}{2}\right]$	[34,56]
35	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}-x\right]$	[35,91]
36	$\left[\frac{1}{4},\ x,\ \frac{1}{2}\right]$	[36,53]
37	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[37,86]
38	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[38,85]
39	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[39,88]
40	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[40,87]
41	$\left[\frac{3}{4},\ x+\frac{1}{2},\ 0\right]$	[41,72]
42	$\left[\frac{1}{2},\ \frac{3}{4},\ x\right]$	[42,81]
43	$\left[\frac{1}{2},\ \frac{3}{4},\ -x\right]$	[43,83]
44	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ 0\right]$	[44,70]
45	$\left[\frac{1}{2},\ \frac{1}{4},\ x\right]$	[45,78]
46	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ 0\right]$	[46,68]
47	$\left[\frac{1}{2},\ \frac{1}{4},\ -x\right]$	[47,79]
48	$\left[\frac{1}{4},\ x+\frac{1}{2},\ 0\right]$	[48,65]

Table 8: Wyckoff site: 48h, site symmetry: 2'..

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ 0\right]$	[1,62]

continued ...

Table 8

No.	position	mapping
2	$\left[x,\ \frac{3}{4},\ 0\right]$	[2,61]
3	$\left[-x,\ \frac{1}{4},\ 0\right]$	[3,64]
4	$\left[-x,\ \frac{3}{4},\ 0\right]$	[4,63]
5	$\left[0,\ x,\ \frac{1}{4}\right]$	[5,60]
6	$\left[\frac{1}{4},\ 0,\ x\right]$	[6,81]
7	$\left[\frac{3}{4},\ 0,\ -x\right]$	[7,83]
8	$\left[0,\ -x,\ \frac{1}{4}\right]$	[8,58]
9	$\left[\frac{3}{4},\ 0,\ x\right]$	[9,78]
10	$\left[0,\ -x,\ \frac{3}{4}\right]$	[10,56]
11	$\left[\frac{1}{4},\ 0,\ -x\right]$	[11,79]
12	$\left[0,\ x,\ \frac{3}{4}\right]$	[12,53]
13	$\left[x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[13,50]
14	$\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[14,49]
15	$\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[15,52]
16	$\left[-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[16,51]
17	$\left[0,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[17,72]
18	$\left[\frac{1}{4},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[18,93]
19	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[19,95]
20	$\left[0,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[20,70]
21	$\left[\frac{3}{4},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[21,90]
22	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[22,68]
23	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[23,91]
24	$\left[0,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[24,65]
25	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}\right]$	[25,86]
26	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$	[26,85]
27	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[27,88]
28	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[28,87]
29	$\left[\frac{1}{2},\ x,\ \frac{3}{4}\right]$	[29,84]
30	$\left[\frac{3}{4},\ 0,\ x+\frac{1}{2}\right]$	[30,57]
31	$\left[\frac{1}{4},\ 0,\ \frac{1}{2}-x\right]$	[31,59]
32	$\left[\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[32,82]
33	$\left[\frac{1}{4},\ 0,\ x+\frac{1}{2}\right]$	[33,54]
34	$\left[\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[34,80]
35	$\left[\frac{3}{4},\ 0,\ \frac{1}{2}-x\right]$	[35,55]
36	$\left[\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[36,77]
37	$\left[x+\frac{1}{2},\ \frac{3}{4},\ 0\right]$	[37,74]
38	$\left[x+\frac{1}{2},\ \frac{1}{4},\ 0\right]$	[38,73]
39	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ 0\right]$	[39,76]
40	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$	[40,75]
41	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[41,96]
42	$\left[\frac{3}{4},\ \frac{1}{2},\ x\right]$	[42,69]
43	$\left[\frac{1}{4},\ \frac{1}{2},\ -x\right]$	[43,71]
44	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[44,94]
45	$\left[\frac{1}{4},\ \frac{1}{2},\ x\right]$	[45,66]
46	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[46,92]
47	$\left[\frac{3}{4},\ \frac{1}{2},\ -x\right]$	[47,67]
48	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[48,89]

Table 9: Wyckoff site: `48i`, site symmetry: `2..`

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[1,14]`
2	$\left[x, \frac{3}{4}, \frac{3}{4}\right]$	`[2,13]`
3	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	`[3,16]`
4	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	`[4,15]`
5	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[5,36]`
6	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	`[6,45]`
7	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	`[7,47]`
8	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	`[8,34]`
9	$\left[\frac{3}{4}, \frac{3}{4}, x\right]$	`[9,42]`
10	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	`[10,32]`
11	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	`[11,43]`
12	$\left[\frac{3}{4}, x, \frac{3}{4}\right]$	`[12,29]`
13	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	`[17,48]`
14	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	`[18,33]`
15	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	`[19,35]`
16	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	`[20,46]`
17	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	`[21,30]`
18	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	`[22,44]`
19	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	`[23,31]`
20	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	`[24,41]`
21	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[25,38]`
22	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[26,37]`
23	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	`[27,40]`
24	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{3}{4}\right]$	`[28,39]`
25	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	`[49,62]`
26	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	`[50,61]`
27	$\left[-x, \frac{1}{4}, \frac{1}{4}\right]$	`[51,64]`
28	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	`[52,63]`
29	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	`[53,84]`
30	$\left[\frac{1}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	`[54,93]`
31	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	`[55,95]`
32	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	`[56,82]`
33	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	`[57,90]`
34	$\left[\frac{1}{4}, -x, \frac{1}{4}\right]$	`[58,80]`
35	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	`[59,91]`
36	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	`[60,77]`
37	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	`[65,96]`
38	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	`[66,81]`
39	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	`[67,83]`
40	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	`[68,94]`
41	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	`[69,78]`
42	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	`[70,92]`
43	$\left[\frac{1}{4}, \frac{1}{4}, -x\right]$	`[71,79]`
44	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	`[72,89]`
45	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	`[73,86]`
46	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[74,85]`

continued ...

Table 9

No.	position	mapping
47	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[75,88]
48	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[76,87]

Table 10: Wyckoff site: 96j, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[x,\ -y,\ -z]$	[2]
3	$[-x,\ y,\ -z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$[z,\ x,\ y]$	[5]
6	$[y,\ z,\ x]$	[6]
7	$[-y,\ z,\ -x]$	[7]
8	$[-z,\ -x,\ y]$	[8]
9	$[-y,\ -z,\ x]$	[9]
10	$[z,\ -x,\ -y]$	[10]
11	$[y,\ -z,\ -x]$	[11]
12	$[-z,\ x,\ -y]$	[12]
13	$\left[x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[13]
14	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[14]
15	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[15]
16	$\left[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[16]
17	$\left[z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	[17]
18	$\left[y,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[18]
19	$\left[-y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$	[19]
20	$\left[-z,\ \frac{1}{2}-x,\ y+\frac{1}{2}\right]$	[20]
21	$\left[-y,\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$	[21]
22	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[22]
23	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[23]
24	$\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	[24]
25	$\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[25]
26	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[27]
28	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[28]
29	$\left[z+\frac{1}{2},\ x,\ y+\frac{1}{2}\right]$	[29]
30	$\left[y+\frac{1}{2},\ z,\ x+\frac{1}{2}\right]$	[30]
31	$\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$	[31]
32	$\left[\frac{1}{2}-z,\ -x,\ y+\frac{1}{2}\right]$	[32]
33	$\left[\frac{1}{2}-y,\ -z,\ x+\frac{1}{2}\right]$	[33]
34	$\left[z+\frac{1}{2},\ -x,\ \frac{1}{2}-y\right]$	[34]
35	$\left[y+\frac{1}{2},\ -z,\ \frac{1}{2}-x\right]$	[35]
36	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$	[36]
37	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[37]
38	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[38]
39	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[39]

continued ...

Table 10

No.	position	mapping
40	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[40]
41	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ y\right]$	[41]
42	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ x\right]$	[42]
43	$\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ -x\right]$	[43]
44	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$	[44]
45	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$	[45]
46	$\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ -y\right]$	[46]
47	$\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ -x\right]$	[47]
48	$\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ -y\right]$	[48]
49	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[49]
50	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[50]
51	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	[51]
52	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[52]
53	$\left[z,\ x,\ y+\frac{1}{2}\right]$	[53]
54	$\left[y,\ z,\ x+\frac{1}{2}\right]$	[54]
55	$\left[-y,\ z,\ \frac{1}{2}-x\right]$	[55]
56	$\left[-z,\ -x,\ y+\frac{1}{2}\right]$	[56]
57	$\left[-y,\ -z,\ x+\frac{1}{2}\right]$	[57]
58	$\left[z,\ -x,\ \frac{1}{2}-y\right]$	[58]
59	$\left[y,\ -z,\ \frac{1}{2}-x\right]$	[59]
60	$\left[-z,\ x,\ \frac{1}{2}-y\right]$	[60]
61	$\left[x,\ y+\frac{1}{2},\ z\right]$	[61]
62	$\left[x,\ \frac{1}{2}-y,\ -z\right]$	[62]
63	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	[63]
64	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	[64]
65	$\left[z,\ x+\frac{1}{2},\ y\right]$	[65]
66	$\left[y,\ z+\frac{1}{2},\ x\right]$	[66]
67	$\left[-y,\ z+\frac{1}{2},\ -x\right]$	[67]
68	$\left[-z,\ \frac{1}{2}-x,\ y\right]$	[68]
69	$\left[-y,\ \frac{1}{2}-z,\ x\right]$	[69]
70	$\left[z,\ \frac{1}{2}-x,\ -y\right]$	[70]
71	$\left[y,\ \frac{1}{2}-z,\ -x\right]$	[71]
72	$\left[-z,\ x+\frac{1}{2},\ -y\right]$	[72]
73	$\left[x+\frac{1}{2},\ y,\ z\right]$	[73]
74	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	[74]
75	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	[75]
76	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	[76]
77	$\left[z+\frac{1}{2},\ x,\ y\right]$	[77]
78	$\left[y+\frac{1}{2},\ z,\ x\right]$	[78]
79	$\left[\frac{1}{2}-y,\ z,\ -x\right]$	[79]
80	$\left[\frac{1}{2}-z,\ -x,\ y\right]$	[80]
81	$\left[\frac{1}{2}-y,\ -z,\ x\right]$	[81]
82	$\left[z+\frac{1}{2},\ -x,\ -y\right]$	[82]
83	$\left[y+\frac{1}{2},\ -z,\ -x\right]$	[83]
84	$\left[\frac{1}{2}-z,\ x,\ -y\right]$	[84]
85	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[85]
86	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[86]

continued ...

Table 10

No.	position	mapping
87	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[87]
88	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[88]
89	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$	[89]
90	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[90]
91	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[91]
92	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[92]
93	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[93]
94	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$	[94]
95	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$	[95]
96	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$	[96]